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(54) **TOUCHLESS IDENTITY CARD EMULATOR SYSTEMS AND METHODS**

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H04W 4/80 (2018.01)
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4/80 (2018.02); **H04W 12/06** (2013.01);
H04W 12/08 (2013.01)

(57)

ABSTRACT

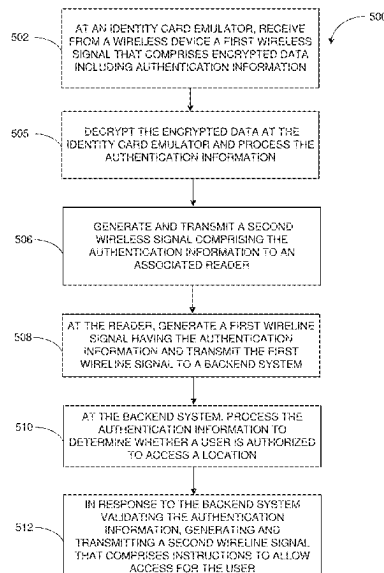
Presented are user-friendly battery powered touchless identity card emulator systems and methods that allow existing ID management installations, such as physical card reader systems, to securely operate without requiring a physical key and irrespective of type, model, shape, and size of reader and card format. Various embodiments integrate wireless functionality to existing systems to enable mobile access to provide advanced user/identity management capabilities for access control systems.

(58) **Field of Classification Search**

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19/0723; H04W 4/80; H04W 12/06;
H04W 12/08; G07C 9/00182

See application file for complete search history.

20 Claims, 6 Drawing Sheets



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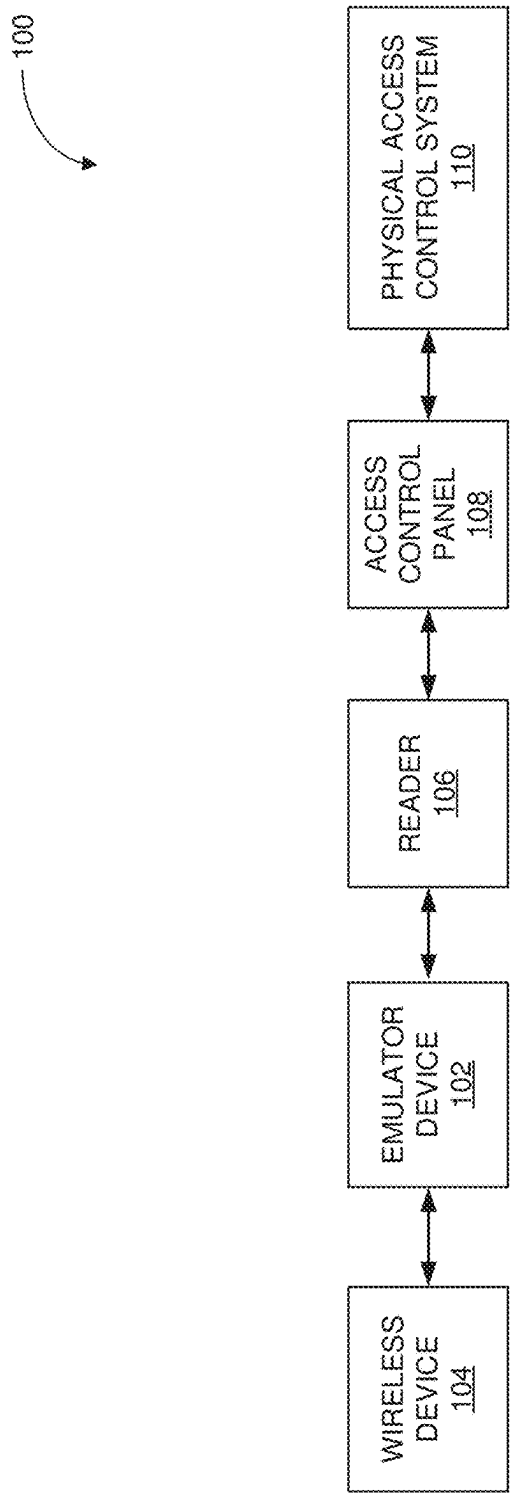


FIG. 1

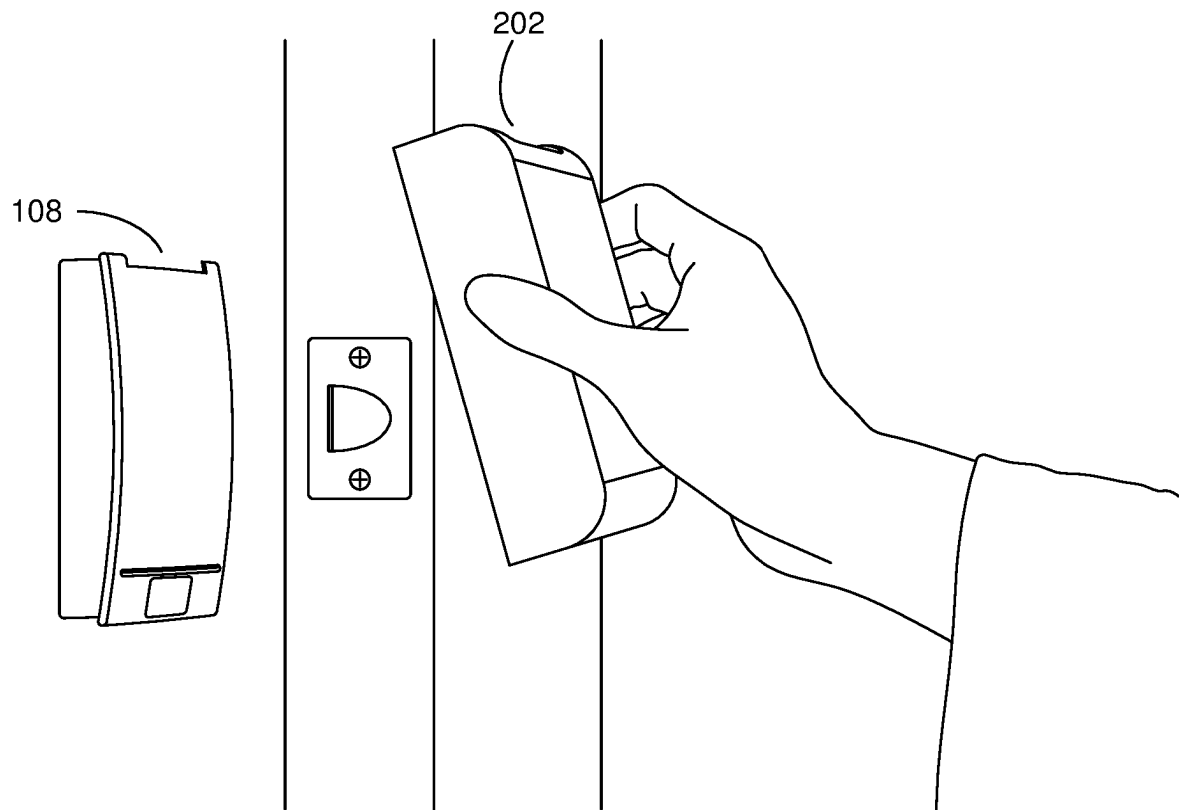


FIG. 2

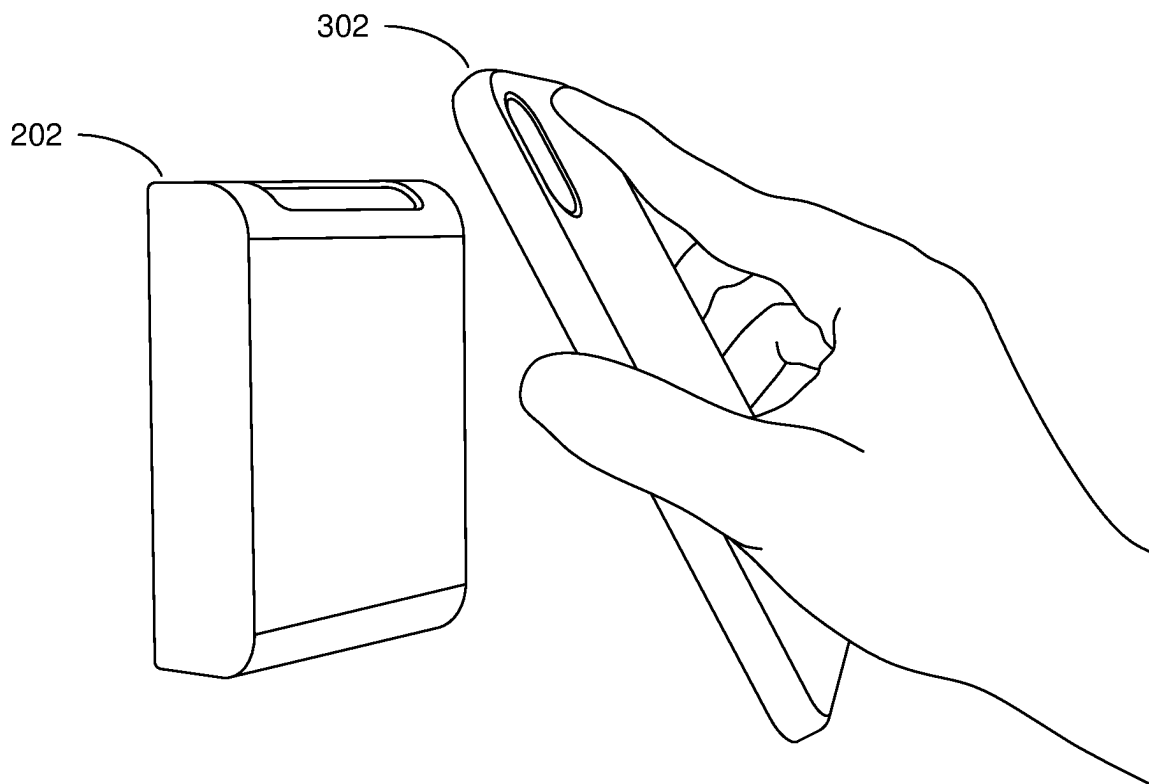


FIG. 3

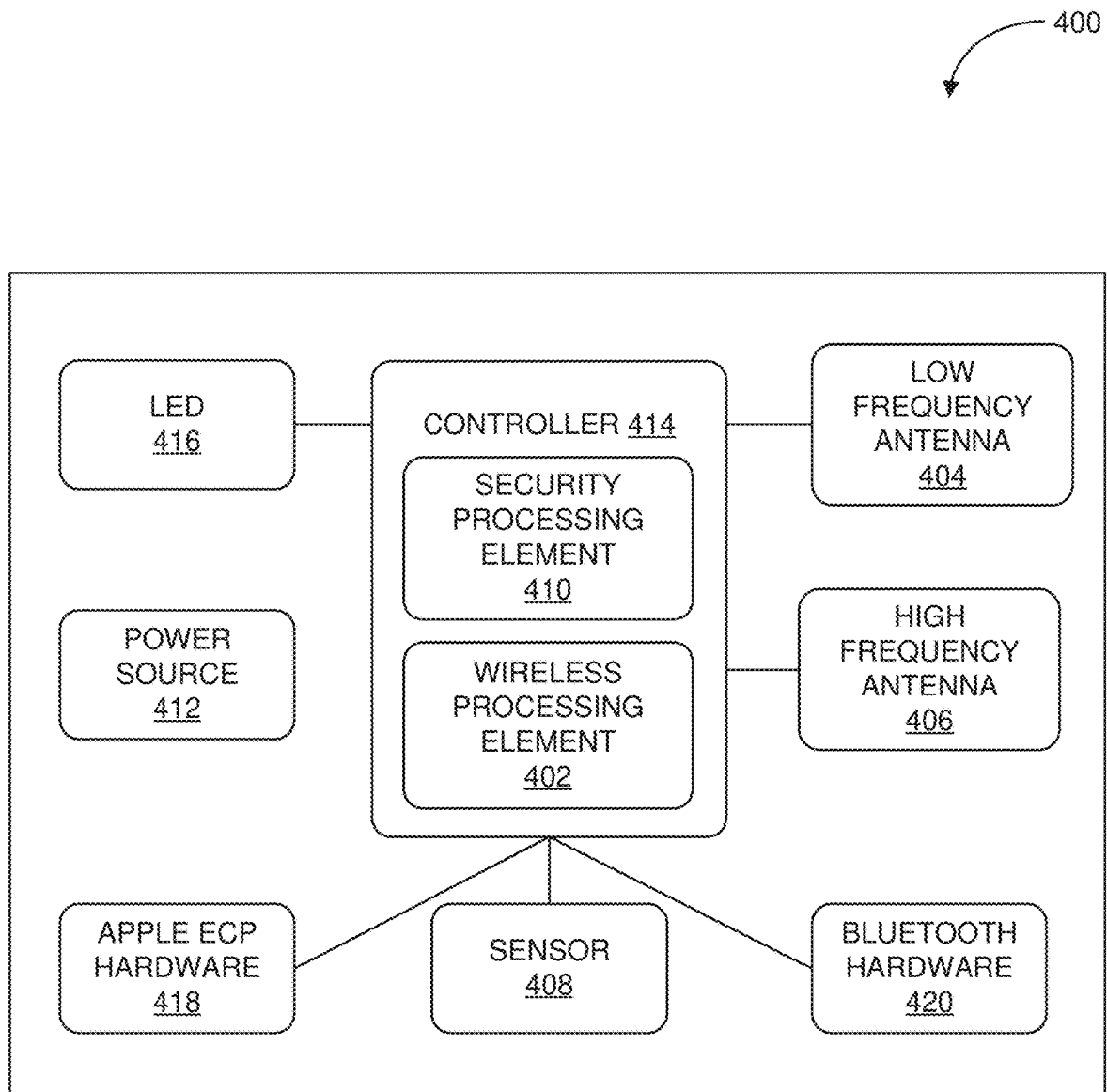


FIG. 4

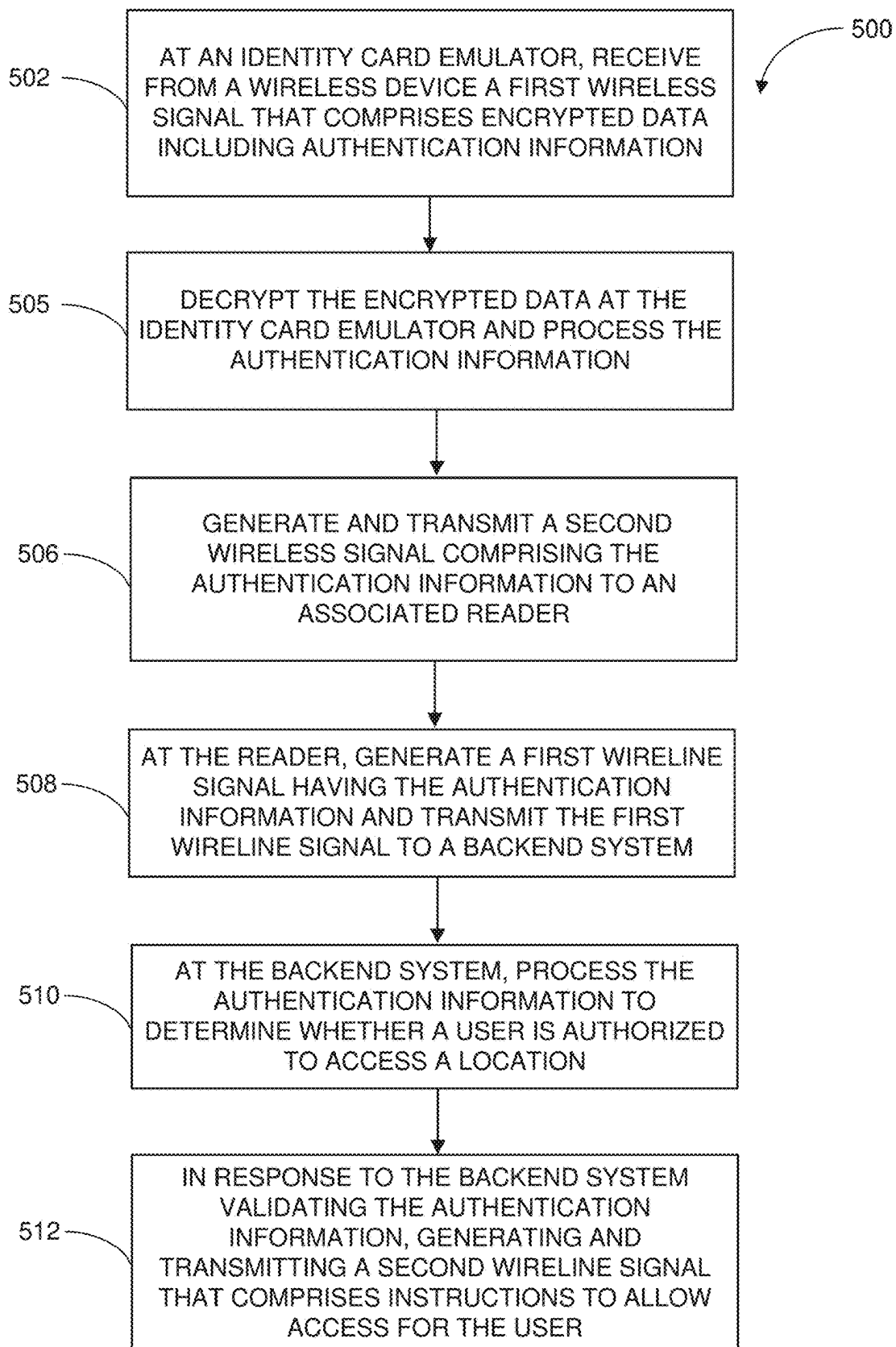


FIG. 5

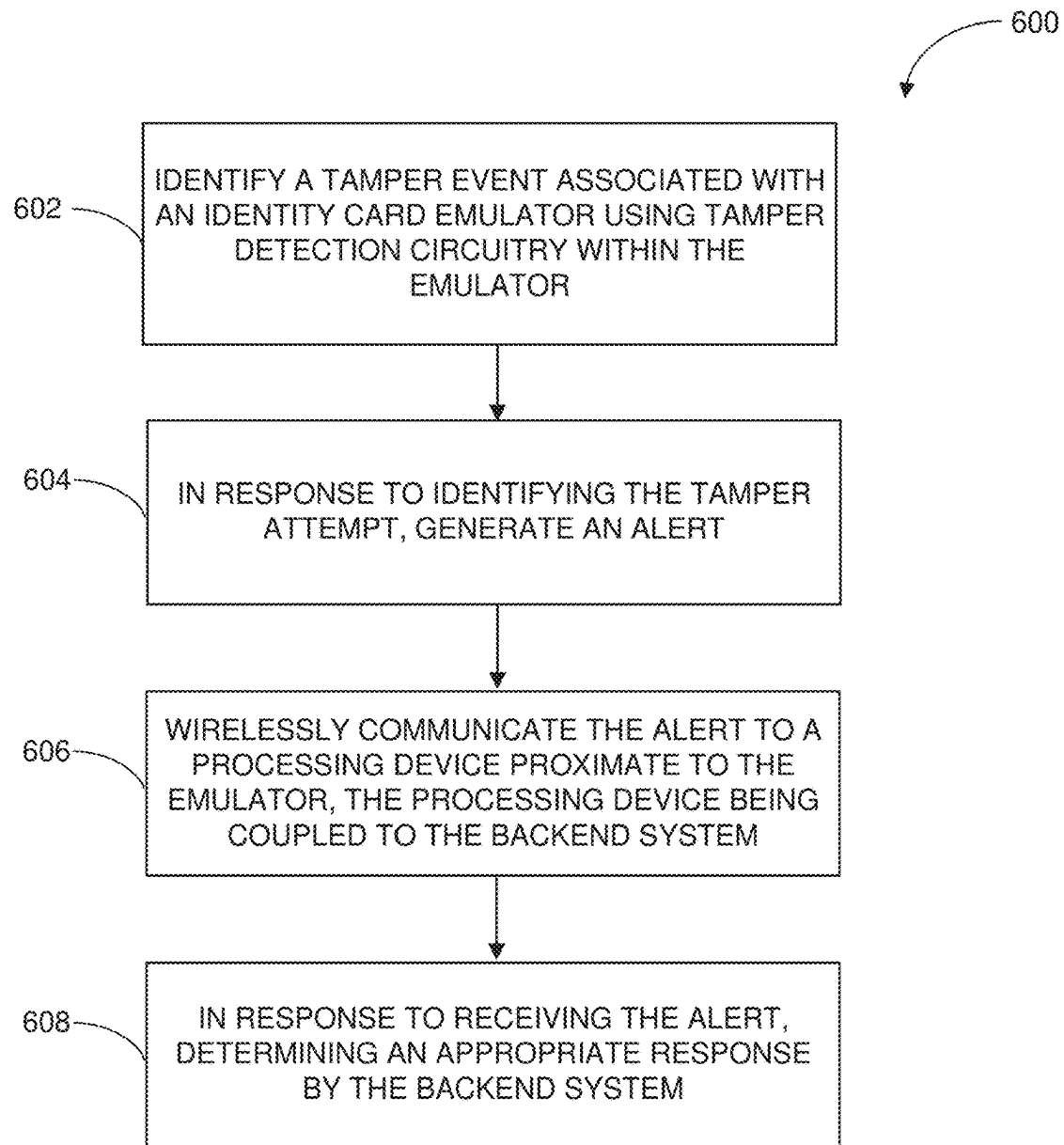


FIG.6

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TOUCHLESS IDENTITY CARD EMULATOR SYSTEMS AND METHODS

BACKGROUND

A. Technical Field

The present disclosure relates generally to identity card systems and methods for access control.

B. Background

RFID card readers that operate with physical ID cards for establishing credentials or physical identity for access control, e.g., as physical badges, are ubiquitous. In addition to being bulky and easily misplaced or lost, physical badges suffer from various security-related drawbacks, including being prone to copying by malicious actors that may use various known sniffing methods to compromise the integrity of an entire identity card system.

Some approaches that attempt to mitigate the disadvantages of physical badges require a complete redesign of existing card readers that are typically installed in a hard-wired fashion. Further, replacing existing access control infrastructure, e.g., to accommodate upgrades that add functionality and/or security features, is a time-consuming, labor-intensive, and costly undertaking that requires rewiring by professional installers, which is exacerbated by different card types and reader types within the same access control system and the need to issue new badges to a large group or users, e.g., the entire workforce of a company.

Accordingly, it is desirable to have low-cost solutions that can upgrade and retrofit existing card reader installations and, ideally, operate securely without the use of a physical key to increase user-friendliness of ID management infrastructure.

BRIEF DESCRIPTION OF THE DRAWINGS

References will be made to embodiments of the invention, examples of which may be illustrated in the accompanying figures. These figures are intended to be illustrative, not limiting. Although the invention is generally described in the context of these embodiments, it should be understood that the teachings of the present disclosure may be extended to other applications that may benefit from functionality enabled herein. For example, as used herein, "access" is not be limited to physical access control systems, such as doors or entrances that require that physical ID is established prior to access being granted. The teachings of the present disclosure may be extended from RFID-type card systems that operate in conjunction with a reader or scanner, such as hotel access cards, transit cards, time clock cards, elevator cards, store coupons, loyalty cards, payment cards, and the like, to other applications not expressly listed herein, such as access to computing system, for example, in elevated security environments, such as in a hospital.

FIG. 1 depicts a typical environment for an emulator device according to various embodiments of the present disclosure.

FIG. 2 illustrates a retrofitting device for use on an existing access control panel, according to various embodiments of the present disclosure.

FIG. 3 illustrates an exemplary use of a wireless device for communicating with a retrofitting device according to various embodiments of the present disclosure.

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FIG. 4 illustrates an exemplary emulator device according to various embodiments of the present disclosure.

FIG. 5 is a flowchart of an illustrative access control process according to various embodiments of the present disclosure.

FIG. 6 is a flowchart of an illustrative process detecting and responding to a tamper attempt associated with an access control system according to various embodiments of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

In the following description, for purposes of explanation, specific details are set forth in order to provide an understanding of the invention. It will be apparent, however, to one skilled in the art that the invention can be practiced without these details. Furthermore, one skilled in the art will recognize that embodiments of the present invention, described below, may be implemented in a variety of ways, such as a process, an apparatus, a system, a device, or a method on a tangible computer-readable medium.

Components, or modules, shown in diagrams are illustrative of exemplary embodiments of the invention and are meant to avoid obscuring the invention. It shall also be understood that throughout this discussion that components may be described as separate functional units, which may comprise sub-units, but those skilled in the art will recognize that various components, or portions thereof, may be divided into separate components or may be integrated together, including integrated within a single system or component. It should be noted that functions or operations discussed herein may be implemented as components. Components may be implemented in software, hardware, or a combination thereof.

Furthermore, connections between components or systems within the figures are not intended to be limited to direct connections. Rather, data between these components may be modified, re-formatted, or otherwise changed by intermediary components. Also, additional or fewer connections may be used. It shall also be noted that the terms "coupled," "connected," or "communicatively coupled" shall be understood to include direct connections, indirect connections through one or more intermediary devices, and wireless connections.

Reference in the specification to "one embodiment," "preferred embodiment," "an embodiment," or "embodiments" means that a particular feature, structure, characteristic, or function described in connection with the embodiment is included in at least one embodiment of the invention and may be in more than one embodiment. Also, the appearances of the above-noted phrases in various places in the specification are not necessarily all referring to the same embodiment or embodiments.

The use of certain terms in various places in the specification is for illustration and should not be construed as limiting. A service, function, or resource is not limited to a single service, function, or resource; usage of these terms may refer to a grouping of related services, functions, or resources, which may be distributed or aggregated.

The terms "include," "including," "comprise," and "comprising" or any of their variants shall be understood to be open terms, and any lists of items that follow are examples and not meant to be limited to the listed items. Any headings used herein are for organizational purposes only and shall not be used to limit the scope of the description or the claims. Each reference mentioned in this patent document is incorporate by reference herein in its entirety.

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It is noted that embodiments described herein are given in the context of readers, but one skilled in the art shall recognize that the teachings of the present disclosure are not limited to reader applications and may equally be used in other contexts.

In this document “card” and “badge” may be used interchangeably. Similarly, the terms “card reader” and “reader;” “identity card” and “access card;” “emulator,” “emulator device,” and “identity card emulator;” and the terms “phone,” “phone hardware,” “app” and “user software” may be used interchangeably. “Reader,” recognized by one of skill in the art, refers to any device that wirelessly obtains user credentials, e.g., an RFID card reader. “User” refers to an end-user who uses one or more embodiments to activate an unlock function.

FIG. 1 depicts a typical environment for an emulator device according to various embodiments of the present disclosure. Environment 100 comprises emulator device 102, wireless communication device 104, reader 106, which may be located in access control panel 108, and physical access control system 110, which may be a cloud-based access control system that uses the internet. Wireless device 104 is any communication device that is capable of wirelessly communicating with other devices, e.g., a smart phone that leverages Bluetooth or any other communication protocol or format to communicate with emulator device 102.

Emulator device 102 is a device according to one or more embodiments herein, for example, emulator device 400 shown in FIG. 4. It is understood that emulator device 102 may be removably attached to an enclosure (not shown) that itself may be removably attached to a housing (e.g., the housing of retrofitting device 202 shown in FIG. 2) that, in embodiments, may be used to at least partially encapsulate access control panel 108. The housing may be a wall-mountable (e.g., door-mounted) device (e.g., a snap-on or more permanent fixture) that may be constructed from any combination of materials. In embodiments, the housing may comprise a weatherproof hardware cover designed to withstand exposure to harsh outdoor conditions. A suitable cover may have particular size that is customized to accommodate one or more different types of access control panels 108 that each itself may have a different size and be associated with a certain type(s) of physical cards.

Further, housing may comprise one or more security features, such as tamper-resistance mechanism to protect against removal of an installed emulator device 102. A suitable mechanism may comprise a set of motion sensors, such as an accelerometer, to detect indicia of tampering, e.g., an attempt to remove emulator device 102. In response, the mechanism may generate an alert. In embodiments, depending on the nature of an alert, emulator device 102 and/or reader 106 may be disabled to protect against unwanted access.

Access control panel 108 may comprise any reader 106 known in the art, e.g., a physical card reader for reading high-frequency (HF) radio frequency (RF) cards (e.g., MIFARE Classic, MIFARE DESFire EV1/EV2/EV3; Apple ECP 2.0, iCLASS, and iCLASS Seos card formats) or low-frequency (LF) RF cards (e.g., HID® Prox Indala AWID, GE Capi, and Honeywell card formats). Physical access control system 110, which may be hardwired to access control panel 108, may use a translation library to verify the validity of the user’s credential prior to determining whether to grant the user access. Once physical access control system 110 determines to grant access, it may use access control panel 108 to initiate an unlocking mechanism. Physical access control system 110 may be used to match

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user credentials, such as a badge ID, group ID, expiration dates, and other information, which may have been paired or mapped to each other or a user profile. In embodiments, a user may pair wireless device 104 directly with emulator device 102, as discussed further below.

It is understood that environment 100 may comprise other and/or additional components not shown in FIG. 1. It is further understood that embodiments herein are not limited to operation in any particular frequency or frequency spectrum, e.g., ISM bands. For example, wireless device 104 may use a wireless communications link and appropriate communication and/or conversion protocols, including any security protocols known in the art, to communicate with any other device, e.g., over an internet connection.

In operation, emulator device 102 may act as a translator that converts a signal comprising user’s credentials that are obtained from wireless device 104, e.g., in an encrypted format, and wirelessly transmits to reader 106 the converted signal in a format that reader 106 can understand. In embodiments, by generating and transmitting to reader 106 a signal similar to that which a physical badge would transmit (e.g., an RF signal comprising a badge ID and/or other content), emulator device 102, in effect, may emulate a physical badge. In embodiments, emulator device 102 may be implemented as a touchless Bluetooth Low Energy (BLE)-to-RFID identity card emulator that, in response to detecting a wireless signal that comprises badge information, generates and transmits to reader 106 a signal that emulates a badge signal that imitates a signal that is typically transmitted to reader 106 by a physical badge. As an example, wireless device 104 which, advantageously, may be implemented in a hardware-agnostic fashion, may obtain a user’s credentials, such as identification information and wirelessly communicate such credentials, e.g., in Bluetooth format, to emulator device 102.

In embodiments, wireless device 104 may automatically connect to emulator device 102 and vice versa, e.g., in response to emulator device 102 detecting that wireless device 104 is placed within a certain range of emulator device 102. In embodiments, emulator device 102 may comprise components that can measure an RSSI value and interpret such value satisfying a condition (e.g., a threshold) as an unlock request (or challenge) that commences an unlock procedure. Once emulator device 102 detects wireless device 104 in this manner, emulator device 102 may automatically establish a connection, e.g., over a secure communication channel, with wireless device 104 without user input. Once a communication channel has been established, wireless device 104 may communicate to emulator device 102 the user’s encrypted credentials, e.g., the user’s badge ID needed to unlock a door.

In embodiments, a user input, such as a tap on a button in an app on a touch screen, may be required to establish a connection between wireless device 104 and emulator device 102. In embodiments, wireless device 104 may receive (e.g., periodically once every few seconds) signals transmitted by emulator device 102 to establish a secure communication channel between wireless device 104 and emulator device 102, e.g., according to a handshake protocol. In embodiments, rather than a user tapping a button on an app or placing the wireless device 104 next to emulator device 102, which may make repeated attempts to wirelessly communicate to activate an unlock procedure, a user may wave their hand at a distance within a certain range near a sensor (not shown in FIG. 1) located within emulator device 102 to trigger a sensor signal. Such a sensor signal may be interpreted by emulator device 102 as a request for access or

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an intent to unlock a door. In embodiments, the signal may be used to commence a communication between wireless device **104** and emulator device **102**, as discussed in greater detail with reference to FIG. **4**.

In embodiments, e.g., in a preliminary or setup phase, a user may cause wireless device **104** to download user-related data (e.g., through an API) and import that data (e.g., in a spreadsheet format) from a backend processing system. As a result, once a user activates the app on wireless device **104**, or waves their hand in front of a sensor, a door may unlock. For example, in an initial setup step, wireless device **104** may download one or more pre-mapped user credentials to authorize a user. Further, the user may set an app in reading mode to scan a physical ID card. The card information may then be transferred to the app, e.g., for storage in a user profile. The app may further be used to authenticate the identity of the user or the user's mobile device requesting access to determine access control and perform initial authorization steps. It is noted that for wireless device **104** implemented as a smart phone an app may run in the background, e.g., while the smart phone is in a locked condition.

Once emulator device **102** obtains a user's credentials, e.g., along with additional information, emulator device **102** may communicate the credentials to card reader **106** in a format that conforms to a format acceptable to card reader **106**. In embodiments, the user's identity or the user's mobile device that acts as a proxy for the user, may be authenticated by performing steps comprising verifying a password, QR code, fingerprint, device code, or location. Authentication may further involve using facial recognition or a video feed, 3-way authentication, or any combination thereof, e.g., depending on a desired level of security. It is understood that at least some authentication steps may be performed with the aid of a remote processor, e.g., by using physical access control system **110** to match phone data with video data. Once a user of wireless device **104** is authenticated, access control may comprise a determination of the user's access rights and verification of access limits based on location (e.g., high-security areas), time (e.g., access expiration dates), credential-based access restrictions, and conditional access (e.g., vaccination status).

In embodiments, authorization may comprise the generation of a confidence score, e.g., based on the presence of an anomaly that may be based on, e.g., tracking data or any other data that allows an identification of the user. Advantageously, there is no need to store and manage any confidential information, such as customer data, which may be stored and handled remotely. Emulator device **102**, e.g., upon authorizing a user, may communicate with reader **106**, e.g., using an LF or HF signal having a specific format, a request to unlock a door. In embodiments, emulator device **102** may communicate, e.g., via Bluetooth or BLE, with a remote device (not shown) that, in response to detecting a potential (e.g., indicia of a tamper event) or actual security breach, may cause emulator device **102** to employ a security mechanism. An exemplary security mechanism may comprise destroying or modifying an encryption key and/or triggering an alarm to perform some predefined action.

It is understood that emulator device **102** may communicate its operational status (e.g., battery charge level), any type of configuration data (e.g., location information), or any other information to the app. Emulator device **102** may do so, for example, in periodic time intervals, based on user traffic or at user-selectable intervals. In embodiments, the remote device may be used to perform a remote unlock operation. In addition, remote device may run a rule-based

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access control software, e.g., to act as a video intercom that utilizes facial recognition and/or computer vision. Matching credentials in this manner is useful when facilitating a network unlock, e.g., in scenarios in which a user forgot or lost wireless device **104**.

Additional uses of the remote device comprise utilizing a thermal camera to perform a temperature scan, facilitating a QR code service, and the like. It is understood that any combination of the remote device, wireless device **104**, and emulator device **102** may be used to obtain other useful information, such as an occupancy count, serve as a visitor management tool (e.g., to validate reservations and track attendance) or increase system security (e.g., by sending tailgating alerts).

FIG. **2** illustrates a retrofitting device **202** for use on an existing access control panel **108**, according to various embodiments of the present disclosure. FIG. **3** illustrates an exemplary use of wireless device **302** for communicating with retrofitting device **202** according to various embodiments of the present disclosure.

FIG. **4** illustrates an exemplary emulator device according to various embodiments of the present disclosure. In embodiments, emulator device **400** comprises power source **412**, wireless processing element **402**, one or more antennae **404**, **406**, sensor(s) **408**, security processing element **410**, controller **414**, LED **416**, Apple ECP hardware **418**, and Bluetooth hardware **420**. Wireless processing element **402** may comprise Bluetooth hardware **420**, e.g., a BLE processor that may operate at the 2.400 GHz-2.4835 GHz frequency range and emit beacon signals as well as transmit BLE advertising packets.

In operation, wireless processing element **402** may be used to establish a secure wireless connection to an end-user's mobile device, such as wireless device **104** shown in FIG. **1**, to exchange encrypted data. The encrypted data, which may comprise user credentials, such as a card ID associated with an RFID card, may be decrypted by security processing element **410** to recover the user's credentials, e.g., in plain text.

In embodiments, security processing element **410** may provide the decrypted data, e.g., after it has been processed by controller **414**, to wireless processing element **402** that may wirelessly transmit the credentials to a reader or scanner. Security processing element **410** may be implemented as an authentication module that comprises memory to store an encryption key. In embodiments, security processing element **410** may comprise any state machine known in the art. As depicted in FIG. **4**, security processing element **410** may be embedded in controller **414**, which may directly or indirectly communicate with and control any number of components in emulator device **400**. It is understood that any component of emulator device **400** may be a standalone component or integrated with one or more other components.

In embodiments, wireless processing element **402** may convert the decrypted data into a format typically generated by an RFID card and read by a reader, such as reader **106** in FIG. **1** to emulate a desired RFID card. The converted data may then be transmitted to the reader, e.g., using one or more antennae. In embodiments, antenna **404** depicted in FIG. **4** may be implemented as a 125 kHz LF antenna, and antenna **406** may be implemented as a 13.56 MHz HF antenna. In operation, antenna **404** may receive, e.g., in a read mode, authentication data from an LF proximity 125 kHz card and communicate at least some of the received data, e.g., over a Bluetooth connection, to a mobile app. Conversely, antenna **404** may transmit the authentication data, e.g., in a write

mode, to a reader device. Similarly, antenna 406 may, in a read mode, receive authentication data from a 13.56 MHz HF MIFARE NFC card and communicate the received data over a Bluetooth connection to the mobile app. Antenna 406 may then transmit authentication data to a reader device.

In embodiments, sensor(s) 408 may be implemented as a proximity sensor (or intent sensor). In embodiments, sensor 408 may detect a gesture, such as a user waving in front of sensor 408, and interpret the gesture as an intent to request access. In embodiments, advantageously, to conserve energy, a gesture may be used to determine when to commence a wireless communication between wireless processing element 402 and an external wireless device. In embodiments, sensor 408 may communicate with controller 414 or wireless processing element 402 to cause a signal to increase BLE advertising packets to commence an unlock procedure. Additionally, emulator device 400 may use sensor 408 to obtain card information from a reader or a card, e.g., in an initialization process that acquires card information related to a user. Alternatively, that information may be obtained from a backend processing system.

In embodiments, power source 412, which may power any number of components in emulator device 400, may be implemented as a set of batteries, e.g., a battery that is removable attached within an enclosure, which, as previous mentioned, may be removably attached to a housing. One skilled in the art will appreciate that power source 412 may be implemented as a stack(s) of cells or a number of interconnected batteries. In embodiments, a battery pack may serve as a backup battery and may be placed external to emulator device 400 within the housing. The battery pack may be used to extend the lifetime of the internal battery. However, this is not intended as a limitation on the scope of the present disclosure since it is understood that a battery pack may be used to power emulator device 400, e.g., to preserve the internal battery for use as a backup battery. In embodiments, a user may install a battery pack by placing it at a pre-defined location within the enclosure and connecting the battery pack to emulator device 400, e.g., via a micro-USB connection.

In embodiments, each of one or more LEDs 416 may be optically coupled to a light pipe or waveguide that may extend from a location near LED 416 to a surface of the enclosure that houses emulator device 400. In operation, LED 416 may provide optical feedback, e.g., to serve as a power on/off indicator. In embodiments, LED 416 may be used as a basic debugging tool, for example, to communicate to an administrator an error or an operation status of emulator device 400, such as a successful pairing, a BLE connection, or other status information.

It is understood that emulator device 400 may comprise any number and combination of additional or different components. For example, emulator device 400 may comprise a microcontroller, memory, additional or different antennae, interfaces, a factory reset button, components to enable reading, writing, and communicating data according to any known wired or wireless communication protocol, e.g., at different spectra, power levels, and levels of encryption, if any.

FIG. 5 is a flowchart of an illustrative access control process according to various embodiments of the present disclosure. In embodiments, access control process may 500 start at step 502 when an identity card emulator receives from a wireless device a first wireless signal that comprises encrypted data associated with authentication information.

At step 504, the identity card emulator decrypts the encrypted data to access and process at least some of the authentication information.

At step 506, the identity card emulator may communicate the authentication information in a second wireless signal to a reader that, at step 508, generates a first wireline signal that comprises the authentication information and transmits that signal to a backend system.

At step 510, the backend system processes the authentication information to determine whether a user is authorized to access a location associated with the card reader.

At step 512, in response to the backend system verifying validating the authentication information, generates and transmits to the reader a second wireline signal that comprises an instruction to grant access to the user.

FIG. 6 is a flowchart of an illustrative process detecting and responding to a tamper event associated with an access control system according to various embodiments of the present disclosure. In embodiments, process 600 starts at step 602 when, a tamper detection circuitry within an identity card emulator identifies a tamper event associated with the identity card emulator. At step 604, in response to identifying the tamper attempt, the identity card emulator generates an alert. At step 606, the identity card emulator wirelessly communicates the alert to a processing device proximate to the emulator, the processing device being coupled to a backend processing system. Finally, at step 608, in response to receiving the alert, the backend processing system determines an appropriate response. One skilled in the art will recognize that: (1) certain steps herein may optionally be performed; (2) steps may not be limited to the specific order set forth herein; (3) certain steps may be performed in different orders; and (4) certain steps may be done concurrently.

In embodiments, the wireless communication device may obtain authentication information in the form of encrypted user credentials, e.g., in a setup or pairing process, and validate the user's credentials. Validated credentials may be used to pair the wireless communication device with the identity card emulator. Pairing may involve the wireless communication device obtaining a unique encryption key to encrypt the user's credentials, e.g., the user's badge/card number in a binary format. This enables secure communication between the wireless communication device and the device that will receive the user's credentials in an encrypted format.

In embodiments, an administrator may log into an administrator software, e.g., a mobile phone app to commence a pairing process. Pairing may begin when the administrator software establishes a communication with a nearby device, e.g., to register the device ID with an online account. Such an account may be serviced at by a backend access control system that prevents unauthorized use of the device, e.g., by communicating a security certificate for the secure transmission of the data. After performing, for example, a firmware update, the administrator software may transmit configuration data to and from the device to enable the device to communicate with one or more readers, e.g., in a format that is compatible with each reader. Exemplary configuration data may comprise reader information, facility codes, security certificates or decryption keys, including default keys, battery modes (e.g., eco), preferred modes of operation (long-range vs. short-range), location data, desired communication frequencies, etc., that may be stored, e.g., in the firmware of the device.

In embodiments, a wireless communication device may establish a secure communication with the identity card

emulator, e.g., in response to receiving a user input and/or detecting a wireless signal and context data to determine a risk level associated with an access request and/or a user. One or more authentication methods and/or risk levels may be indicated to the user, e.g., via the wireless communication device and, based on the risk level, the user may be authenticated. The wireless communication device may wirelessly communicate user credentials as encrypted data to the identity card emulator, e.g., to activate an unlocking mechanism.

One skilled in the art will recognize no computing system or programming language is critical to the practice of the present invention. It will be appreciated to those skilled in the art that the preceding examples and embodiments are exemplary and not limiting to the scope of the present disclosure. It is intended that all permutations, enhancements, equivalents, combinations, and improvements thereto that are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the true spirit and scope of the present disclosure. It shall also be noted that elements of any claims may be arranged differently including having multiple dependencies, configurations, and combinations.

What is claimed is:

1. An access control system comprising:
 - a wireless device that generates a first wireless signal; an identity card emulator communicatively coupled with the wireless device, the identity card emulator comprising:
 - a wireless processing element communicatively coupled to the wireless device to receive the first wireless signal, the first wireless signal comprising encrypted data associated with authentication information;
 - a security processing element coupled to the wireless processing element, the security processing element performing steps comprising using a decryption key to decrypt the encrypted signal to access the authentication information, the wireless processing element performing steps comprising:
 - generating a second wireless signal that comprises the authentication information and
 - transmitting the second wireless signal;
 - a reader communicatively coupled to the identity card emulator, the reader, in response to receiving the second wireless signal, performing steps comprising:
 - generating a first wireline signal comprising the authentication information; and
 - transmitting the first wireline signal;
 - a backend system communicatively coupled to the reader, the backend system, in response to receiving the first wireline signal, performing steps comprising:
 - processing the authentication information to determine whether a user is authorized to access a location; and
 - in response to determining that the user is authorized to access the location generating a second wireline signal comprising an instruction to grant access to the user; and
 - transmitting the second wireline signal to the reader to initiate an unlocking mechanism.
2. The access control system according to claim 1, wherein the security processing element comprises a state machine that decrypts the data.
3. The access control system according to claim 1, wherein the identity card emulator is removably attached to a housing.

4. The access control system according to claim 3, wherein the housing comprises one or more sensors to detect a physical activity associated with the identity card emulator.

5. The access control system according to claim 4, wherein the one or more sensors comprise at least one of a location sensor or a vibration sensor to detect a tampering event.

6. The access control system according to claim 1, wherein the unlocking mechanism facilitates access to a physical location.

7. An identity card emulator comprising:

- a wireless processing element that receives from a wireless device a first wireless signal that comprises encrypted data that comprises authentication information;
- a decryption module coupled to the wireless processing element, the decryption module, in response to receiving the encrypted data, decrypts the encrypted signal to access the authentication information; and
- a controller coupled to the decryption module, the controller causing the wireless processing element to generate a second wireless signal that communicates the authentication information to a reader to initiate an unlocking process.

8. The identity card emulator according to claim 7, wherein the authentication information authenticates at least one of the wireless device or a user of the wireless device.

9. The identity card emulator according to claim 7, further comprising one or more antennae coupled to wireless processing element, the one or more antennae generating at least one of the first wireless signal or the second wireless signal.

10. The identity card emulator according to claim 7, further comprising a sensor configured to detect a gesture that is indicative of an access request.

11. The identity card emulator according to claim 10, wherein the sensor is a proximity sensor.

12. The identity card emulator according to claim 7, further comprising a light emitting diode that is optically coupled to at least one of a light pipe or waveguide to provide optical feedback.

13. The identity card emulator according to claim 7, wherein wireless processing element is configured to detect an RSSI signal that is indicative of at least one of an unlock request or challenge to commence an unlock procedure.

14. An access control method comprising:

- receiving a first wireless signal at an emulator device, the first wireless signal comprises encrypted data associated with authentication information;
- decrypting, at the emulator device, the encrypted data to access the authentication information;
- generating, at the emulator device, a second wireless signal that comprises the authentication information;
- transmitting the second wireless signal to a reader to initiate an unlocking process; and
- generating, at the reader, a third signal and transmitting the third signal to a backend system that, in response to determining the authentication information associated to an authorized user, grants access to a location.

15. The access control method according to claim 14, further comprising using the first wireless signal to establish a secure communication to a wireless device.

16. The access control method according to claim 15, wherein secure communication is established in response to detecting an RSSI signal.

17. The access control method according to claim **15**, wherein secure communication is established in response to detecting a sensor signal.

18. The access control method according to claim **14**, further comprising assigning a risk level to a context state 5 based on one or more rules associated with the context state.

19. The access control method according to claim **18**, further comprising determining one or more authentication methods based on the risk level.

20. The access control method according to claim **18**, 10 further comprising using a tamper detection circuit to detect a tamper attempt and initiating an action.

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